

Lecture 1: From Assignments to Teams

Aleksi Anttila Marco Degano

Team Semantics: Linguistic and Philosophical Applications
ESSLLI 2026, Prague

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Course Team



Marco Degano



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 - ▶ modality, negation, and compositionality

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- ▶ *Optional* exercises for each class will be provided at the end of each slide deck.

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- ▶ **Friday:** Philosophical applications: compositionality, negation, modality.

Readings

- ▶ The first team semantics paper: W Hodges, Compositional semantics for a language of imperfect information, Logic Journal of the IGPL, Volume 5, Issue 4, July 1997, Pages 539–563, <https://doi.org/10.1093/jigpal/5.4.539>
- ▶ High-level overview: <https://plato.stanford.edu/entries/logic-dependence/>

Outline

1. Motivation

2. Precursors

3. Team Semantics

The Classical Point of Evaluation

- In ordinary propositional semantics, a valuation assigns truth values to propositional variables.

	p	q
$v_{p\bar{q}}$	1	0

- Then we ask whether a formula is true at this single valuation:
 $v_{p\bar{q}} \models p \wedge \neg q$

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- ▶ Using a team as an **information state**: the actual valuation is somewhere in T , but the information state does not determine which row is actual.

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- ▶ This team rules out that $p = 0$ and $q = 1$, but it leaves open three possibilities.

The Space of Possibilities

- For $N = \{p, q\}$, 2^N contains four valuations.

	p	q
v_{pq}	1	1
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The four possible valuations.

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$$T_1 = \{v_{pq}\}$$

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- A team selects **a set of rows**.

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- ▶ $T \models \neg p$ means that every $v \in T$ has $v(p) = 0$.
- ▶ T leaves p **open** if some $v \in T$ has $v(p) = 1$ and some $w \in T$ has $w(p) = 0$.

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- ▶ So, in this sense of dependence, in this team q depends on p .
- ▶ But if we say $v_3(q) = 1$, then dependence is lost.
- ▶ Team semantics can add atoms that express **team-level properties**, like dependence.

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- ▶ Example: ordinary possible-worlds semantics

Possible-Worlds Semantics

- ▶ Ordinary possible-worlds semantics is not team-like in the relevant sense.
- ▶ A proposition is a set of worlds.
- ▶ But satisfaction is evaluated at a single world: $M, w \models \phi$.
- ▶ The set $\llbracket \phi \rrbracket = \{w \in W \mid M, w \models \phi\}$ is a denotation.
- ▶ In a team-setting, the denotation for ϕ would be a set of sets of points.

Discourse Representation Theory

- DRT is only indirectly team-like.

$$K = \left[\begin{array}{c} x \quad y \\ \hline \text{MAN}(x) \\ \text{DONKEY}(y) \\ \text{OWNS}(x, y) \end{array} \right] \quad \rightsquigarrow \quad X_K = \{g \mid M, g \models K\}$$

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- ▶ But the primitive semantic object of DRT is a discourse representation structure, not a team.

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- ▶ Dynamic semantics asks what accepting a sentence does to information.
- ▶ A meaning is therefore a **context-change potential**.
- ▶ Instead of only asking whether ϕ is true at w , we ask how ϕ updates an information state.

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- ▶ More information means fewer live possibilities:

$$s[\phi] \subseteq s$$

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- ▶ The rows preserve dependencies between discourse referents.
- ▶ A formula specifies a transition between assignment states, written $G[\![\phi]\!]H$.

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- ▶ In propositional dynamic semantics, the relevant state is a set of worlds $s \subseteq W$
- ▶ In first-order dynamic semantics, the relevant state may be a set of assignments, or a set of world-assignment pairs.
- ▶ This is team-like because semantic operations act on sets of evaluation points.

Linear Quantification

- ▶ In ordinary first-order logic, quantifiers are **linearly ordered**.
- ▶ Example: $\forall x \exists y \forall z \exists w \phi(x, y, z, w)$.

Linear Quantification

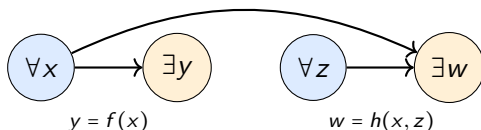
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- ▶ The witness for y may depend on x .
- ▶ The witness for w may depend on x, z
- ▶ Dependence is **implicit in the order** of quantifiers.



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- ▶ Linear prefixes impose a **total order** on quantifier choices.
- ▶ But some dependence patterns are naturally partial.
- ▶ Example: one may want y to depend only on x and w to depend only on z .
- ▶ Ordinary linear order cannot directly express this independence pattern.

Branching Quantifiers

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- ▶ The choice of y does not depend on z .
- ▶ The choice of w does not depend on x .
- ▶ This is not yet team semantics, but it isolates dependence and independence as semantic phenomena.

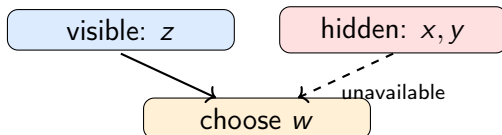


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- ▶ Independence-friendly logic makes **independence restrictions syntactic**.
- ▶ The notation $(\exists w/W)$ means that the choice of w must be independent of the variables in W .
- ▶ The Henkin pattern can be written linearly as $\forall x \exists y \forall z (\exists w / \{x, y\}) \phi(x, y, z, w)$.
- ▶ The slash set $\{x, y\}$ prevents the choice of w from using information about x or about the earlier choice of y .
- ▶ The choice of w may still depend on z .



Informal Game-Theoretical Semantics

Two players: a verifier trying to show the formula ϕ is true, and a falsifier trying to show it is false, with respect to a model $\mathcal{M}$ and assignment s . ϕ is true/false w.r.t. $\mathcal{M}$ and s if the verifier/falsifier has a winning strategy.

- ▶ If ϕ is atomic, the verifier wins if $\mathcal{M} \models_s \phi$. The falsifier wins if $\mathcal{M} \not\models_s \phi$
- ▶ If ϕ is $\psi \vee \chi$, the verifier picks $\theta := \psi$ or $\theta := \chi$. Play the game with θ .
- ▶ If ϕ is $\psi \wedge \chi$, the falsifier picks $\theta := \psi$ or $\theta := \chi$. Play the game with θ .
- ▶ ϕ is $\exists x\psi$: The verifier picks $a \in \mathcal{M}$. Play the game with $s(a/x)$ & ψ .
- ▶ ϕ is $\forall x\psi$: The falsifier picks $a \in \mathcal{M}$. Play the game with $s(a/x)$ & ψ .
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- ▶ If the falsifier chooses $x = x$, the verifier wins because $a = a$.

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With these quantifiers, the game is one of *imperfect information*.

Matching pennies game in independence-friendly logic

Each player has a coin that they secretly turn to heads or tails. The coins are revealed simultaneously. I wins if the coins are both heads or both tails. Otherwise II wins.

Let $\mathcal{M}$ be a model with domain $\{h, t\}$ and ϕ be the sentence

$$\forall x(\exists y/\{x\})x = y$$

Neither player has a winning strategy, so $\mathcal{M} \not\models \phi$ and $\mathcal{M} \not\models \neg\phi$.

II (falsifier) does not have a winning strategy: all they can do is choose the value of x . They cannot guarantee that I will not choose the same value for y .

I (verifier) does not have a winning strategy: II first chooses the value of x . I does not know which value was picked, so they cannot ensure that they pick the same value for y .

Hintikka had a wider philosophical program with IF logic at its centre:

- ▶ Unlike standard Tarskian semantics, the game-theoretical semantics for logic explain *why* the resulting truth relation is a truth relation: game-theoretical truth is justified in terms the actual activities by which we verify or falsify sentences (a language game à la Wittgenstein for truth).

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- ▶ From the viewpoint of game-theoretical semantics, the restriction to only perfect-information games is arbitrary.
- ▶ Given this (and more), IF logic should replace classical first-order logic as the 'fundamental' logic used in mathematics.

Hintikka also argued against the importance commonly placed on compositionality in the semantics of both natural and formal languages. He assumed that it would be impossible to provide a (reasonable) compositional semantics for IF logic, and used this as a point against compositionality.

Barwise, commenting on Hintikka:

Linguistically, the discovery of branching quantification [in natural language] would force us to re-examine, and perhaps re-interpret, Frege's principle of compositionality according to which the meaning of a given expression is determined by the meanings of its constituent phrases. For example, the meaning of a branching quantifier expression of logic

$$\left(\begin{smallmatrix} \forall x & \exists y \\ \forall z & \exists w \end{smallmatrix} \right) A(x, y, z, w)$$

cannot be defined inductively in terms of simpler formulas, by explaining away one quantifier at a time.

Hodges' development of team semantics showed that there was a reasonable compositional semantics for IF.

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- ▶ These sets were called trumps (!). In later terminology, they are teams.
- ▶ The satisfaction relation becomes $M \models_X \alpha$, where X is a set of assignments.
- ▶ This is the first-order analogue of evaluating propositions on information states.

Outline

1. Motivation
2. Precursors
3. Team Semantics

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$T \upharpoonright \{p\}$	p
v_1	1
v_2	1
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Formulas

- Propositional dependence logic PD has formulas

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- ▶ Negation occurs only in literals p and $\neg p$.
- ▶ $\text{Var}(\phi)$ is the finite set of variables occurring in ϕ .
- ▶ Satisfaction $T \models \phi$ is defined whenever $\text{Var}(\phi) \subseteq \text{Dom}(T)$.

Literal Clauses

- ▶ $T \models p$ iff $v(p) = 1$ for every $v \in T$.
- ▶ $T \models \neg p$ iff $v(p) = 0$ for every $v \in T$.
- ▶ Literals are evaluated universally over the team.

T	p	q
v_1	1	1
v_2	1	0

- ▶ $T \models p$
- ▶ $T \not\models q$
- ▶ $T \not\models \neg q$

Dependence Atom

$$T \models \text{dep}(p_1, \dots, p_n, q)$$

iff for all $v, w \in T$,

$$v(p_i) = w(p_i) \text{ for every } i \leq n \implies v(q) = w(q)$$

Equivalently, there is a function

$$f : \{0, 1\}^n \rightarrow \{0, 1\}$$

such that, for every $v \in T$,

$$v(q) = f(v(p_1), \dots, v(p_n))$$

T	p	q
v_1	0	1
v_2	1	0
v_3	1	0

Failure of Dependence

- ▶ Dependence fails when two rows agree on the determining variables but disagree on the determined variable.

T	p	q
v_1	0	0
v_2	0	1
v_3	1	1

- ▶ v_1 and v_2 agree on p .
- ▶ v_1 and v_2 disagree on q .
- ▶ Hence $T \not\models \text{dep}(p, q)$.

Conjunction

- ▶ $T \models \phi \wedge \psi$ iff $T \models \phi$ and $T \models \psi$.
- ▶ Conjunction keeps the same team.

T	p	q
v_1	1	1
v_2	1	1

- ▶ $T \models p$.
- ▶ $T \models q$.
- ▶ Hence $T \models p \wedge q$.

Split Disjunction

- ▶ $T \models \phi \vee \psi$ iff there are $S, U \subseteq T$ such that $S \cup U = T$, $S \models \phi$, and $U \models \psi$.
- ▶ The team is split into a ϕ -part and a ψ -part.
- ▶ Empty subteams are allowed.
- ▶ The subteams may overlap.
- ▶ This is the lax split disjunction clause.

Example: Split Disjunction

- ▶ Let $T = \{v_1, v_2, v_3\}$.

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- ▶ $T \models p \vee \neg p$.

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- ▶ Take $S = \{v_1, v_2\}$ and $U = \{v_3\}$.
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- ▶ $S \cup U = T$.

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Locality

- ▶ Satisfaction depends only on the variables occurring in the formula.
- ▶ If $\text{Var}(\phi) \subseteq \text{Dom}(T)$, then $T \models \phi$ iff $T \upharpoonright \text{Var}(\phi) \models \phi$.
- ▶ If $T \upharpoonright \text{Var}(\phi) = U \upharpoonright \text{Var}(\phi)$, then $T \models \phi$ iff $U \models \phi$.
- ▶ Extra columns are semantically irrelevant.

E.g. for $\text{Var}(\phi) = \{p\}$:

T	p	q
v_1	1	1
v_2	1	0
v_3	0	1

Semantic Consequence

- ▶ For a set of formulas Γ , define $\Gamma \models \phi$ iff for every team T , if $T \models \gamma$ for all $\gamma \in \Gamma$, then $T \models \phi$.
- ▶ A formula ϕ is valid iff $T \models \phi$ for every team T over a domain containing $\text{Var}(\phi)$.
- ▶ By locality, validity may be checked over teams $T \subseteq 2^{\text{Var}(\phi)}$.

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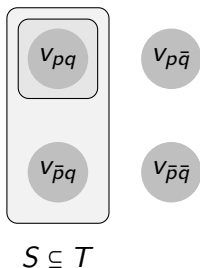
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- ▶ We ask what happens when we remove rows.
- ▶ We ask what happens when we add rows.
- ▶ We ask what happens when we take unions of teams.
- ▶ These properties are useful for understanding definability and the characteristics of meaning in different team logics, as well as for classifying logics: many later logics differ exactly here.

Downward Closure

- ▶ A formula ϕ is downward closed if $T \models \phi$ and $S \subseteq T$ imply $S \models \phi$.
- ▶ Removing rows preserves truth.



- ▶ Example: if $T \models q$, then every subteam $S \subseteq T$ also satisfies q .

Lax and Strict Split

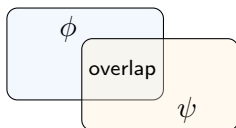
- ▶ In the lax clause for disjunction, S and U **may overlap**.
- ▶ In the strict clause for disjunction, one additionally requires $S \cap U = \emptyset$.
- ▶ For downward closed formulas, lax and strict split are equivalent.

Exercise: Lax vs. Strict Split

Assume ϕ and ψ are downward closed. Show that:

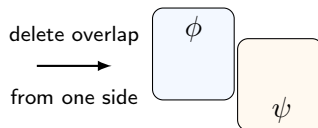
$$T \models_{\text{lax}} \phi \vee \psi \iff T \models_{\text{strict}} \phi \vee \psi$$

lax split



$$S \cup U = T, \text{ possibly } S \cap U \neq \emptyset$$

strict split



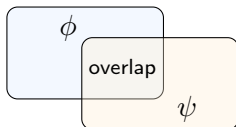
$$S' \cup U' = T, S' \cap U' = \emptyset$$

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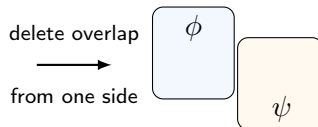
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lax split



$$S \cup U = T, \text{ possibly } S \cap U \neq \emptyset$$

strict split



$$S' \cup U' = T, S' \cap U' = \emptyset$$

By downward closure: $S \models \phi \Rightarrow S' \models \phi$ and $U \models \psi \Rightarrow U' \models \psi$.

Upward Closure

- ▶ A formula ϕ is upward closed if $T \models \phi$ and $T \subseteq S$ imply $S \models \phi$.
- ▶ Adding rows preserves truth.
- ▶ Literals are not upward closed.
- ▶ Dependence atoms are not upward closed.

T	p	S	p
v_1	1	v_1	1
		v_2	0

- ▶ $T \models p$
- ▶ $T \subseteq S$
- ▶ $S \not\models p$

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- ▶ Let ϕ contain no dependence atoms.
- ▶ Then $T \models \phi$ iff $\{v\} \models \phi$ for every $v \in T$ iff $v \models_{\text{cl}} \phi$ for every $v \in T$.

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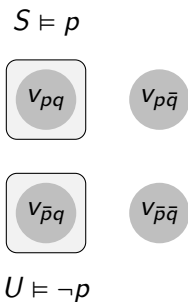
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Flatness of the Classical Fragment

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- ▶ That is, classical formulas are not only flat, but their team semantics coincides with their standard semantics on singletons.
- ▶ Hence the team semantics of ordinary propositional formulas is pointwise classical.

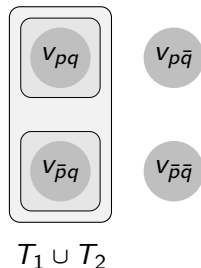
Split Disjunction Is Not Boolean Choice

- ▶ $T \models \phi \vee \psi$ does not generally mean that $T \models \phi$ or $T \models \psi$.
- ▶ It means that T can be covered by a ϕ -subteam and a ψ -subteam.
- ▶ For flat formulas, this agrees with ordinary pointwise disjunction.
- ▶ For non-flat formulas, the split itself can matter.



Union Closure

- ▶ A formula ϕ is union closed if $T_i \models \phi$ for all $i \in I \neq \emptyset$ implies $\bigcup_{i \in I} T_i \models \phi$.
- ▶ Pooling satisfying teams preserves truth.



- ▶ Example: if $T_1 \models q$ and $T_2 \models q$, then $T_1 \cup T_2 \models q$.

Union Closure Can Fail

- ▶ Dependence atoms are not union closed. Why?
- ▶ Singleton teams always satisfy dependence atoms.

T_1	p	q	T_2	p	q	$T_1 \cup T_2$	p	q
v_1	0	0	v_2	0	1	v_1	0	0
						v_2	0	1

- ▶ $T_1 \models \text{dep}(p, q)$ and $T_2 \models \text{dep}(p, q)$.
- ▶ But $T_1 \cup T_2 \not\models \text{dep}(p, q)$.
- ▶ The union creates a disagreement on q among rows that agree on p .

Empty Team Property

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- ▶ Every PD-formula has the empty team property.
- ▶ For literals, this is vacuous universal quantification.
- ▶ For dependence atoms, this is vacuous universal quantification.
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- ▶ For split disjunction? Take both subteams to be $\emptyset$.

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 - ▶ Now suppose ϕ is downward- and union closed, and has the empty team property.
 - ▶ By downward closure, $T \models \phi$ implies $\{v\} \models \phi$ for all $v \in T$.
 - ▶ By union closure (if $T \neq \emptyset$) and the empty team property (if $T = \emptyset$), $\{v\} \models \phi$ for all $v \in T$ implies $T \models \phi$.

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- ▶ Closure properties encode assumptions about the kinds of meanings formulas express.
- ▶ If teams are information states, downward closure says: *once established, a fact remains established when we learn more.*
- ▶ Union closure says that satisfaction is preserved when compatible information states are pooled.
- ▶ Failure of a closure property is also informative: it shows that the logic can express genuinely team-level properties.

First-Order Teams

- ▶ Let M be a first-order model with domain $|M|$.
- ▶ Let V be a finite set of variables.
- ▶ An assignment over V is a function $g : V \rightarrow |M|$.
- ▶ A team over M with domain V is a set $X \subseteq |M|^V$.

From Propositional to First-Order Teams

- ▶ Propositional teams were **sets of valuations**: $T \subseteq 2^N$
- ▶ First-order teams are **sets of assignments**: $X \subseteq |M|^V$
- ▶ The guiding idea is the same.

X	x	y	z
g_1	a	b	c
g_2	a	b	d
g_3	e	f	c

First-Order Literals and Boolean Clauses

- ▶ First-order literals are evaluated pointwise:

$$M, X \models \alpha \iff M, g \models \alpha \text{ for every } g \in X$$

- ▶ Similarly,

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- ▶ Conjunction and disjunction work exactly as in the propositional case.

Quantifiers as Team Operations

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- ▶ If $a \in |M|$, write $g[a/x]$ for the assignment just like g , except that x is mapped to a .
- ▶ Existential quantification **supplements** the team: $M, X \models \exists x \phi$ iff there is a function

$$F : X \rightarrow \mathcal{P}(|M|) \setminus \{\emptyset\} \text{ such that } M, X[F/x] \models \phi$$

where

$$X[F/x] = \{g[a/x] \mid g \in X, a \in F(g)\}$$

Example: Supplementing

- ▶ Let $|M| = \{a, b, c, d\}$.
- ▶ Let $X = \{g_1, g_2\}$.
- ▶ Suppose $F(g_1) = \{a, b\}$ and $F(g_2) = \{b\}$.

X	y	$X[F/x]$	y	x
g_1	c	$g_1[a/x]$	c	a
g_2	d	$g_1[b/x]$	c	b
		$g_2[b/x]$	d	b

Universal Quantification

- ▶ Universal quantification **duplicates** the team:

$$M, X \models \forall x \phi \iff M, X[|M|/x] \models \phi$$

- ▶ Here

$$X[|M|/x] = \{g[a/x] \mid g \in X, a \in |M|\}$$

- ▶ So every assignment receives every possible value for x .

Example: Duplicating

X	y	
g_1	a	$ M = \{a, b\}$
g_2	b	

$X[M /x]$	y	x
$g_1[a/x]$	a	a
$g_1[b/x]$	a	b
$g_2[a/x]$	b	a
$g_2[b/x]$	b	b

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- ▶ The distinction matters for logics that are not downward closed.

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- ▶ First-order dependence atoms have the form

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- ▶ Their truth is defined exactly like propositional dependence:

$$M, X \models \text{dep}(x_1, \dots, x_n, y)$$

iff for all $g, h \in X$,

$$g(x_i) = h(x_i) \text{ for every } i \leq n \quad \Rightarrow \quad g(y) = h(y)$$

- ▶ Thus, inside X , the value of y is **determined by** the values of $x_1, \dots, x_n$.

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- ▶ $\forall x \forall z \exists y (\text{dep}(x, y) \wedge P(x, y))$
- ▶ $\exists f \forall x \forall z (P(x, f(x)))$

Exercise 1: Semantics

Consider the following team T in a model M with domain the set of numbers $\{0, 1, 2, 3\}$:

T	x	y	z
g_1	0	2	3
g_2	0	1	1
g_3	1	2	3
g_4	1	1	0

Determine whether:

- (i) $M, T \models \text{dep}(x, y)$
- (ii) $M, T \models \text{dep}(z, y)$
- (iii) $M, T \models \text{dep}(x, z) \vee \text{dep}(y, z)$
- (iv) $M, T \models \exists q(q \neq x \wedge q \neq y \wedge q \neq z)$
- (v) $M, T \models \forall q(\text{dep}(q, x)) \vee \forall q(\text{dep}(q, y))$

Consider a dependence atom $\text{dep}(\emptyset, y)$ with an empty list of variables $x_1, \dots, x_n$. When is $\text{dep}(\emptyset, y)$ true?

Exercise 2: Closure Properties

Classify each property as downward closed, union closed, upward closed, and/or empty-team valid.

- (i) $T \models p$
- (ii) $T \models \text{dep}(p, q)$
- (iii) T is compatible with p
- (iv) $T = \emptyset$
- (v) $T \subseteq \{v \mid v(p) = 1\}$

For every failed closure property, give a counterexample.

Exercise 3: A new extension

Let $D = \{a, b\}$ and let $X = \{i_1, i_2\}$ with $i_1(y) = a$ and $i_2(y) = b$.

We extend X with a new variable x . Write (c, d) for the assignment with $y = c$ and $x = d$.

	extension Y	allowed?
Y_1	$\{(a, a), (b, a)\}$	yes
Y_2	$\{(a, a), (b, b)\}$	no
Y_3	$\{(a, a), (b, a), (b, b)\}$	no
Y_4	$\{(a, b), (b, a)\}$	no
Y_5	$\{(a, b), (b, b)\}$	yes

	extension Y	allowed?
Y_6	$\{(a, b), (b, a), (b, b)\}$	no
Y_7	$\{(a, a), (a, b), (b, a)\}$	no
Y_8	$\{(a, a), (a, b), (b, b)\}$	no
Y_9	$\{(a, a), (a, b), (b, a), (b, b)\}$	yes

1. What property distinguishes the admissible extensions from the non-admissible ones?
2. Find a general description of the admissible x -extensions of an arbitrary team X . Your description should have the form $Y = \{i[c/x] \mid \dots\}$. Use this extension to define a new quantifier $\exists^{\text{new}}_X$.
3. Can this quantifier be defined using the ordinary lax existential quantifier together with a new atom of the form $\#(x, \vec{y})$?

Exercise 4: Second-order logic and expressible properties

Reformulate the following sentences of dependence logic in ESO, using function quantifiers:

1. $\forall x \forall u \exists y \exists z (\text{dep}(x, y) \wedge \text{dep}(u, z) \wedge (u = x \rightarrow y = z))$
2. $\exists q(\text{dep}(\emptyset, q))$ (Hint: we also allow 0-ary function quantifiers, i.e., constant quantifiers)

What do the following sentences express?

1. $\forall x \forall u \exists y \exists z (\text{dep}(x, y) \wedge \text{dep}(u, z) \wedge (u = x \leftrightarrow y = z))$
2. $\forall x \exists y \exists q (\text{dep}(x, y) \wedge \text{dep}(\emptyset, q) \wedge y \neq q)$

What can we say about a model M that makes the following sentence true?

$$\forall x \forall u \exists y \exists z \exists q (\text{dep}(x, y) \wedge \text{dep}(u, z) \wedge (u = x \leftrightarrow y = z) \wedge \text{dep}(\emptyset, q) \wedge y \neq q)$$